

HW/Lab 6 Sol: MIMO Implementation

ECSE 4560/6560: Modern Communication Systems

RPI Fall 2025

Question 1 Sol: OFDM-MIMO Decoding

1. **Case 1:** (1×2 SIMO)

Decoding rule:

$$h_A^* y_{i,A} + h_B^* y_{i,B} = (|h_A|^2 + |h_B|^2)x_i + h_A^* n_{i,A} + h_B^* n_{i,B} .$$

Effective signal power:

$$P_S = \mathbb{E} \left[(|h_A|^2 + |h_B|^2)x_i^2 \right] = (|h_A|^2 + |h_B|^2)^2 \mathbb{E}[|x_i|^2] = (|h_A|^2 + |h_B|^2)^2 ,$$

where the signal power is normalized as $\mathbb{E}[|x_i|^2] = 1$.

Effective noise power:

$$P_N = \mathbb{E} \left[|h_A^* n_{i,A} + h_B^* n_{i,B}|^2 \right] = |h_A|^2 \mathbb{E}[n_{i,A}^2] + |h_B|^2 \mathbb{E}[n_{i,B}^2] = (|h_A|^2 + |h_B|^2)\sigma^2 .$$

Therefore, the effective SNR is:

$$\frac{P_S}{P_N} = \frac{(|h_A|^2 + |h_B|^2)^2 \mathbb{E}[|x_i|^2]}{(|h_A|^2 + |h_B|^2)\sigma^2} = \frac{(|h_A|^2 + |h_B|^2) \mathbb{E}[|x_i|^2]}{\sigma^2} = \frac{|h_A|^2 + |h_B|^2}{\sigma^2} .$$

2. **Case 2:** (1×2 SIMO)

Decoding rule:

$$\begin{aligned} & h_A^* y_{i,A} + h_B^* y_{i,B} + h_C^* y_{i,C} + h_D^* y_{i,D} \\ &= (|h_A|^2 + |h_B|^2 + |h_C|^2 + |h_D|^2)x_i + h_A^* n_{i,A} + h_B^* n_{i,B} + h_C^* n_{i,C} + h_D^* n_{i,D} . \end{aligned}$$

Similar to Case 1, the effective SNR is:

$$\frac{P_S}{P_N} = \frac{(|h_A|^2 + |h_B|^2 + |h_C|^2 + |h_D|^2) \mathbb{E}[|x_i|^2]}{\sigma^2} = \frac{|h_A|^2 + |h_B|^2 + |h_C|^2 + |h_D|^2}{\sigma^2} ,$$

which is higher (better) than Case 1.

3. **Case 3:** (2×1 MISO)

Precoding rule (normalized):

$$c_{i,A} = \frac{h_A^*}{\sqrt{|h_A|^2 + |h_B|^2}} , \quad c_{i,B} = \frac{h_B^*}{\sqrt{|h_A|^2 + |h_B|^2}} ,$$

which gives

$$y_i = \left(\sqrt{|h_A|^2 + |h_B|^2} \right) x_i + n_i .$$

Effective signal power:

$$P_S = \mathbb{E} \left[\left| \sqrt{|h_A|^2 + |h_B|^2} x_i \right|^2 \right] = (|h_A|^2 + |h_B|^2) \mathbb{E}[|x_i|^2] = |h_A|^2 + |h_B|^2 .$$

Effective noise power:

$$P_N = \mathbb{E}[|n_i|^2] = \sigma^2 .$$

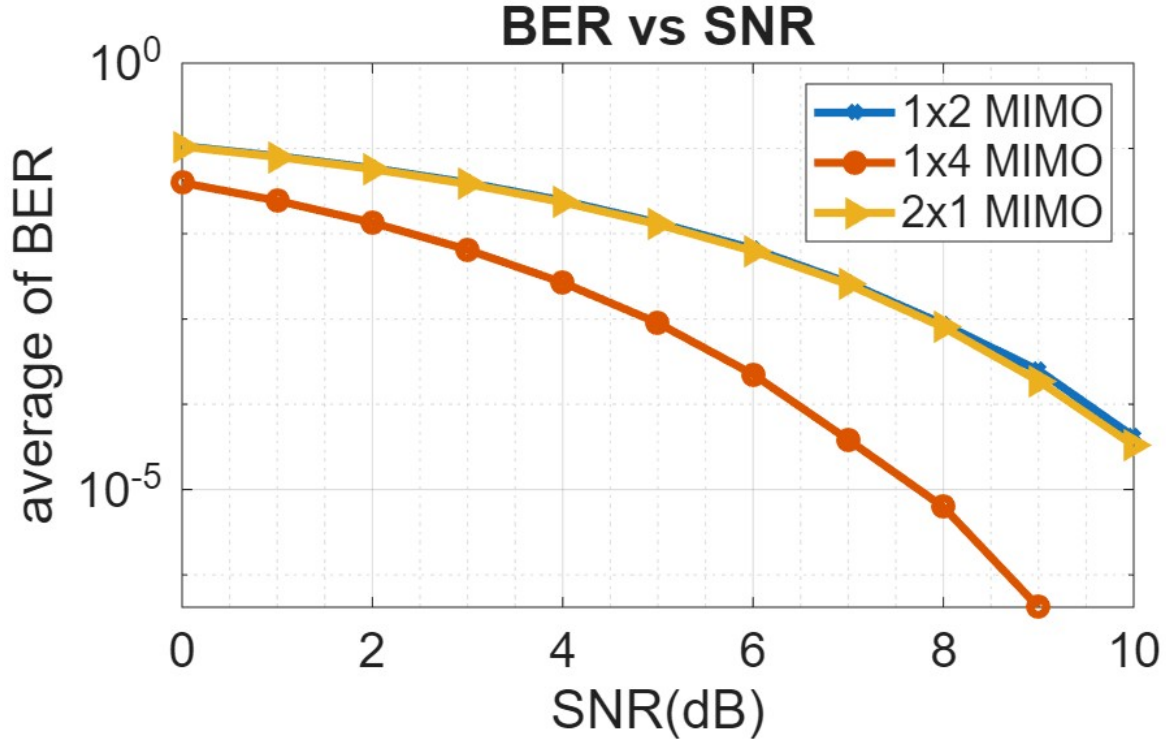
Therefore, the effective SNR is:

$$\frac{P_S}{P_N} = \frac{(|h_A|^2 + |h_B|^2) \mathbb{E}[|x_i|^2]}{\sigma^2} = \frac{|h_A|^2 + |h_B|^2}{\sigma^2} ,$$

which is identical to Case 1, while lower (worse) than Case 2.

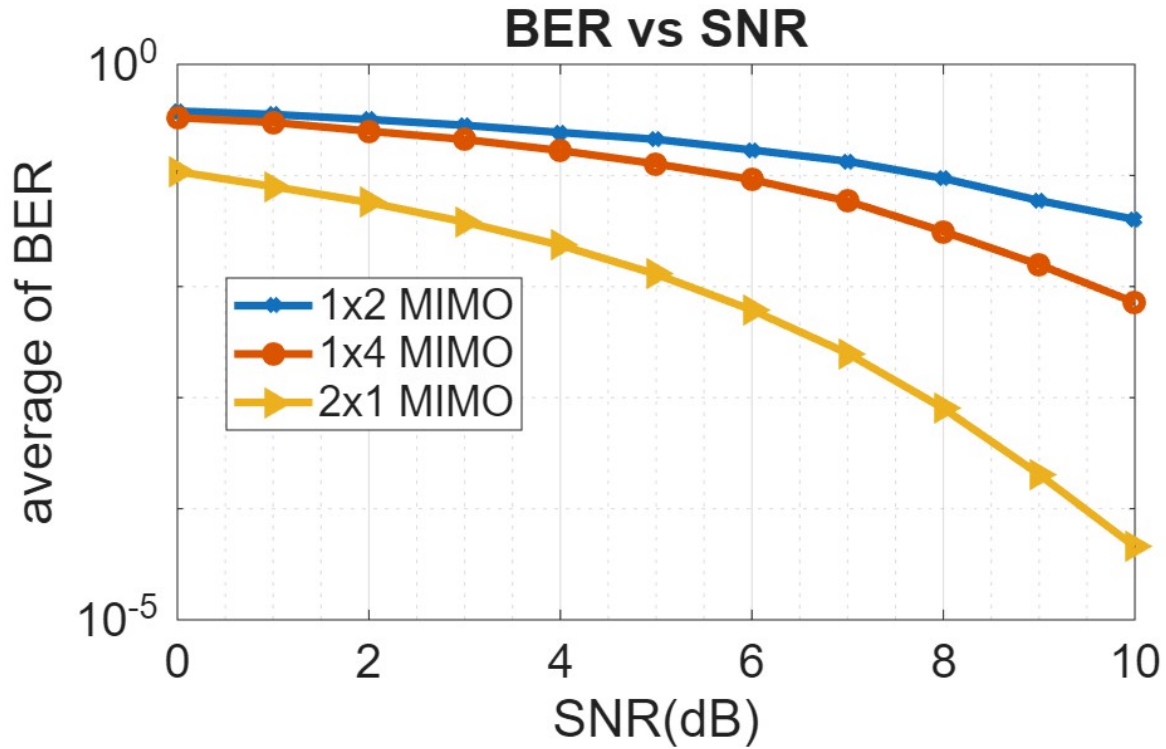
Question 2 Sol: BER-SNR Plots with Explanation

1. BER plot when Case 1 and Case 2 use ground truth CSIR



The BER values are as expected. For the same BER target, Case 1 and Case 3 require equally higher SNR. This is because given the same transmit power and noise power, Case 1 and Case 3 have equally lower effective SNR.

2. (ECSE 6560 only) BER plot when Case 1 and Case 2 use estimated CSIR



Compared to the previous plot, the BER values of Case 1 and Case 2 increase because of channel estimation error.

3. (ECSE 6560 only) How to use channel estimates in Case 3

A simple way would be run an extra iteration at the beginning with ground truth CSIT to obtain the channel estimates for the following iteration. Note that the estimated channels are different for different subcarriers, so the precoding needs to be done for different subcarriers separately.